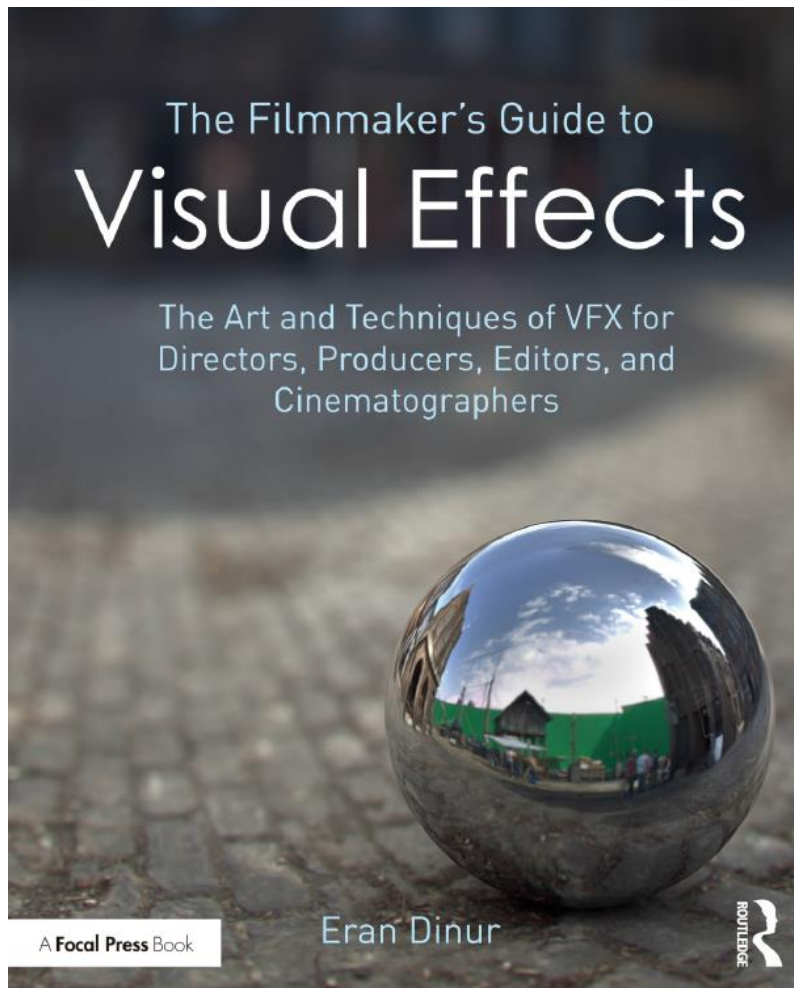


Camera Shots and VFX

Reading

The Filmmaker's Guide to
Visual Effects, Eran Dinur,
2021

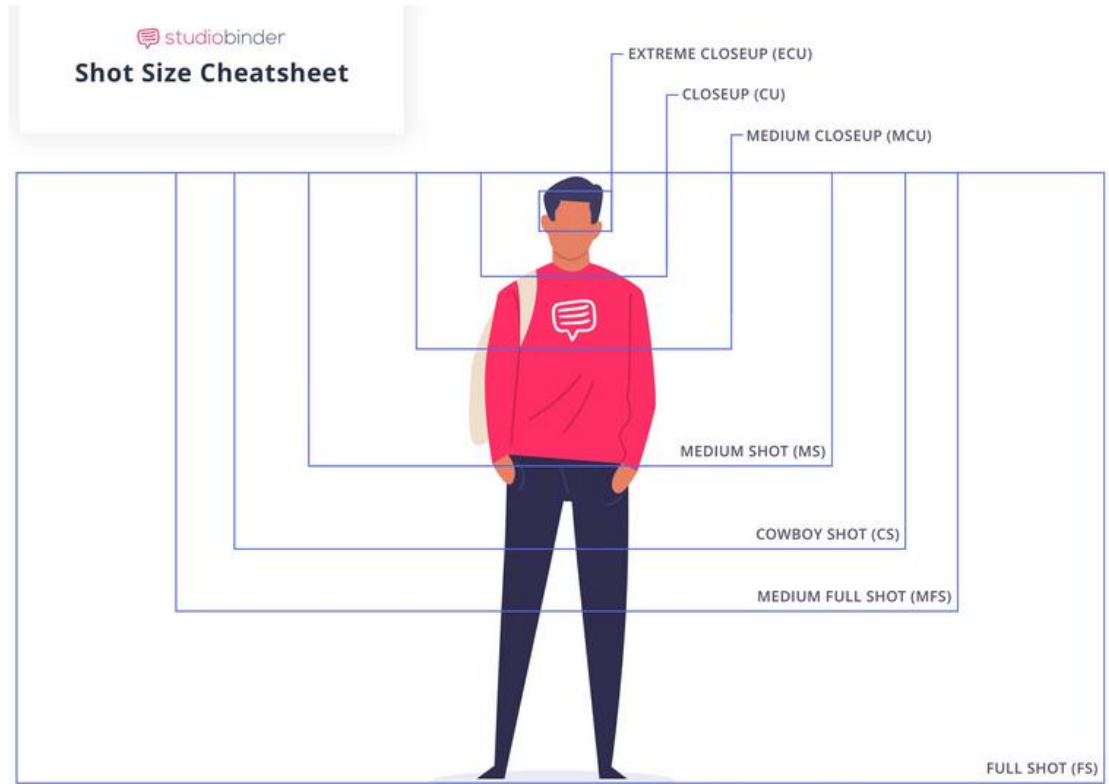


A camera shot

- **A camera shot** is composed of the series of frames that are shot **uninterrupted** from the moment the camera **starts** rolling until it **stops**.
- Camera shots are an essential aspect of filmmaking and video productions, because by combining different types of shots, angles and camera movements, the filmmakers are able to emphasize specific emotions, ideas and movement for each scene.

Camera Shot Size Types

1. [Extreme Wide Shot \(ELS\)](#)
2. [Long Shot \(LS\) / Wide Shot \(WS\)](#)
3. [Full Shot \(FS\)](#)
4. [Medium Long Shot \(MLS\) /
Medium Wide Shot \(MWS\)](#)
5. [Cowboy Shot](#)
6. [Medium Shot \(MS\)](#)
7. [Medium Close Up \(MCU\)](#)
8. [Close Up \(CU\)](#)
9. [Extreme Close Up \(ECU\)](#)
10. [Establishing Shot](#)



1- Extreme Long Shot (ELS) or Extreme Wide Shot (EWS)

An **extreme long shot** (or extreme wide shot) make **your subject appear small against their location** .

يظهر الهدف صغيراً وسط المكان

You can use an extreme long shot to make your subject feel distant or unfamiliar **لاظهار الهدف بعيد أو غريب عن المكان**.

Take a look at this extreme wide shot from one of the **best action movies** *Mad Max: Fury Road*

It can also make your subject feel **overwhelmed** by its location. Of all the various camera shots out there, consider using the extreme long shot when you need to emphasize the location or **isolation**.

يعطي احساس بالانعزال أو الضالة



(EWS) Extreme Wide Shot Example in *Mad Max: Fury Road*

<https://www.studiobinder.com/blog/ultimate-guide-to-camera-shots/>

2-Long Shot (LS) or Wide Shot (WS)

The [long shot](#) (also known as a wide shot, abbreviated “WS”) is the same idea, but a **bit closer**.

If your subject is a person then his or her **whole body** will be in view — but **not filling the shot**.

Here's an example of the wide shot size from one of the [best 2015 movies](#) *The Martian*

The wide shot also lets us see the **beautiful background imagery**, as well as the onlookers which will make any big moment more cinematic. لإظهار البيئة المحيطة.



(WS) Wide Shot Example in *The Martian*

3- Full Shot (FS)

Now let's talk about camera shots that let **your subject fill the frame** (usually in Height) يظهر طول الشخصية كاملاً ويملاً المشهد (usually in Height) while keeping emphasis on scenery المشهد.

Of all the different types of camera shots in film, full shots can be used to feature multiple characters in a single shot, like this full shot size example from Marvel's Guardians of the Galaxy:



(FS) Full Shot Example in *Django Unchained*



(FS) Full Shot Example in *Guardians of the Galaxy*

4-Medium Wide Shot (MWS)

A [medium long shot](#) (aka medium long shot) frames the subject from roughly the **knees up**. It splits the difference between a full shot and a medium shot.



(MWS) Medium Wide Camera Shot Example in The Usual Suspects

5- Cowboy Shot

A variation on this is the [Cowboy Shot](#), which frames the subject from roughly **mid-thighs up**. It's called a "cowboy shot" because it is used in Westerns to frame a gunslinger's gun or holster on his hip

من منتصف الشخصية) أو منتصف الفخذ (للأعلى



(CS) Cowboy Shot Example in *The King's Speech*

6- Medium Shot (MS)

Let's move onto camera shots that reveal your subject in more detail. (this type and the next types also)

The [medium shot](#) is one of the most common camera shots. It's similar to the cowboy shot above, but frames from roughly the **waist up and through the torso**

. من الخصر لأعلى ومن خلال الجذع

So it emphasizes more of your subject while keeping their surroundings visible.

Medium shots may seem like the most standard camera shot around, but every shot size you choose will have an effect on the viewer. A medium shot can often be used in **dialogue scenes** that have an important moment later that will be shown in a close-up shot.

تستخدم بكثرة في المشاهد الحوارية



(MS) Medium Shot Example in *X-Men: Days of Future Past*

7- Medium Close Up Shot (MCU)

The [medium close-up](#) frames your subject from roughly the **chest up** من الصدر. So it typically favors the face, but still keeps the subject somewhat distant.

The medium close-up camera shot size keeps the characters distant even during their face-to-face conversation. يمكن استخدامه أيضاً خلال الحوارات.



(MCU) Medium Close Up Shot Example in [No Country for Old Men](#)

8- Close Up (CU)

You know it's time for a [close-up shot](#) when you want to reveal a **subject's emotions and reactions**. The close-up camera shot fills your frame with a part of your subject. If your subject is a person, it is **often their face**.



(CU) Close-Up Shot Example in The Usual Suspects

9- Extreme Close Up (ECU)

An extreme close-up is the most you can fill a frame with your subject. It often shows eyes, mouth and gun triggers. In extreme close-up shots, smaller objects get great detail and are the focal point.

It can be also use to show details of objects or tools



(ECU) Extreme Close-Up Camera Shot Example in X-Men: First Class

Shots indicating camera angle/placement

زاوية أو مكان الكاميرا

In addition to subject size within a frame, shot types can also indicate where a camera is placed in relation to the subject. Here are some commonly used terms:

1- Eye Level Shot taken with the camera approximately at **human eye level**, resulting in a neutral effect on the audience. مستوى النظر العادي



Shots indicating camera angle/placement

2- **High Angle** Subject is photographed from above eye level. This can have the effect of making the subject seem weak or frightened.

تعطي تأثيراً بالضعف



Shots indicating camera angle/placement

3- **Low Angle** Subject is photographed from below eye level. This can have the effect of making the subject look **powerful, heroic, or dangerous**. احساس بالخطر أو البطولة



Shots indicating camera angle/placement

4- **Dutch Angle/Tilt** Shot in which the camera is set at an angle on its roll axis so that the horizon line is not level. It is often used to show a disoriented or uneasy psychological state.

يكون الكادر مائلًا وغير مواز لمستوى الأرض مما يعطي إحساساً بعدم الاتزان



Shots indicating camera angle/placement

5-Over-the-Shoulder Shot A popular shot where a subject is shot from behind the shoulder of another, framing the subject anywhere from a Medium to Close-Up.

The shoulder, neck, and/or back of the head of the subject facing away from the camera remains viewable, making the shot useful for showing reactions during conversations.

It tends to place more of an emphasis on the connection between two speakers rather than the detachment or isolation that results from single shots.



Shots indicating camera angle/placement

6- **Bird's-Eye View** (aka Top Shot) A high-angle shot that's taken from directly overhead and from a distance. The shot gives the audience a wider view and is useful for showing direction and that the subject is moving, to highlight special relations, or reveal to the audience elements outside the boundaries of the character's awareness. The shot is often taken from on a crane or helicopter/ Surveillance camera. زاوي أشبه بكاميرات المراقبة



Shots indicating camera angle/placement

7- Point of View Shot (POV) Shot intended to mimic what a particular character in a scene is seeing.

محاكاة زاوية رؤية شخص

This puts the audience directly into the head of the character, letting them experience their emotional state. Common examples are of a character waking up, drifting into unconsciousness **فقدان الوعي**, or looking through a scope or binoculars.



Shots indicating camera angle/placement

8- Knee Level Shot

This is when your camera height is about as low as your subject's knees.

They can emphasize a character's superiority/power, if paired with a low angle.



*Shot Types: Knee Level Shot in **Home Alone***

VFX (visual effects)

- Visual Effects (VFX) involves the **integration** of **live action footage** مشهد واقعي (which may include in-camera special effects) and **generated imagery** (digital or optics, animals or creatures) which look realistic, but would be **dangerous, expensive, impractical, time-consuming or impossible to capture on film.**
- Visual effects using computer-generated imagery (CGI) have more recently become accessible to the independent filmmaker with the introduction of affordable and relatively easy-to-use animation and compositing software.

VFX Uses-1

- 1-To add impressive content to the movie



VFX Uses-2

- 2-To create an imaginary environment



VFX Uses-3

- 3-To add special effects to a movie

BEFORE



AFTER



Common VFX Softwares

- Adobe After Effects
- Maxon Cinema 4D
- Autodesk Maya
- Syntheyes
- 3Ds Max
- Houdini
- Boujou
- Mocha
- Nuke
- Natron (Open source)

Ref <https://www.nfi.edu/vfx/>

Open Source Visual Effects Software

1. Blender
2. Blackmagic Fusion
3. Natron
4. The Academy Color Encoding System (ACES)
5. Gaffer
6. Alembic
7. Pixar USD

<https://inspirationtuts.com/open-source-visual-effects-software/>

Difference between VFX & 3D Animation

- **Some references** make difference between 3D Animation and Visual Effects
- 3D Animation traditionally uses **100% CG (computer graphics)** elements.
- Although techniques are similar, the Visual Effects industry relies heavily on **capturing real world footage** (plates) and integrates CG elements into them seamlessly.
- Example:
 - “Toy Story” is 3D Animation industry,
 - “The Avengers” is Visual Effects.

Visual Effects vs. Special Effects

Visual effects	Special effects
VFX requires a computer for implementation.	SFX encompasses elements like pyrotechnics, fake rain, or prosthetic makeup. الألعاب النارية ، المطر المزيف ، أو المكياج التعويضي
Visual effects do not take place during the actual shoot. لا تحدث أثناء التصوير الفعلي	Special effects are realized on set. تحدث في موقع تصوير
Visual effects create, enhance or manipulate images for film or other media.	Special effects are made using a model or animatronics on a location set.
Examples of visual effect •Dragons flying through the sky in Game of Thrones •A spaceship flying through space in Star Wars.	Examples of special effects •Purposeful and controlled explosions •Fake gunshot wounds
VFX are considered digital effects as they enhance the quality of visual storytelling.	SFX are considered <u>practical effects</u> , as they are functional elements that feature while the movie plot unfolds.

VFX types

- 1- Computer-generated imagery (CGI)
- 2- Compositing and Green screen VFX
- 3- Motion capture visual effects

1- Computer-generated imagery (CGI):

- CGI is the most common visual effect used. Artists use **3D modelling** objects to create objects or surfaces. And those 3D models are digitized into a computer.
- **Actions** Created by Computer **without** performing it into reality.
- Visual effects can mean any kind of effect that was is shot directly in the camera. So it could mean creating specific lighting. Or, it might be enhancing an image.

2- Compositing and Green screen VFX:

- Compositing **تركيب المشاهد** is a VFX visual technique that combines multiple images into one. This technique is called “**Chroma** keying.”
- Compositing requires **live-action** **تنفيذ فعلي للحركة** to be shot in front of a green or blue screen.
- **Green screens** are solid in color and can be **replaced** with a new background image. But that doesn't happen until after post-production **يحدث الاستبدال بعد انتهاء التصوير**.
- In this way, compositing and green screen VFX occur both during and after production. So the process is created with live capture and digital enhancement.
- These techniques can create an other-worldly scene. And the technical prowess is what makes such scenes so realistic.

3- Motion capture visual effects:

- Motion capture, or **mocap**, combines live action with CGI VFX.
- Motion detection can be by:
 - Computer vision
 - Special Marks on scene.
 - Sensors
- The film records the body movement of the actor. And if it captures their **facial** expressions تعبيرات الوجه as well, it is called performance الممثل capture .
- Those movements are then transferred to a digital model. Here, the VFX artist can use the actor's work to enhance the scene.

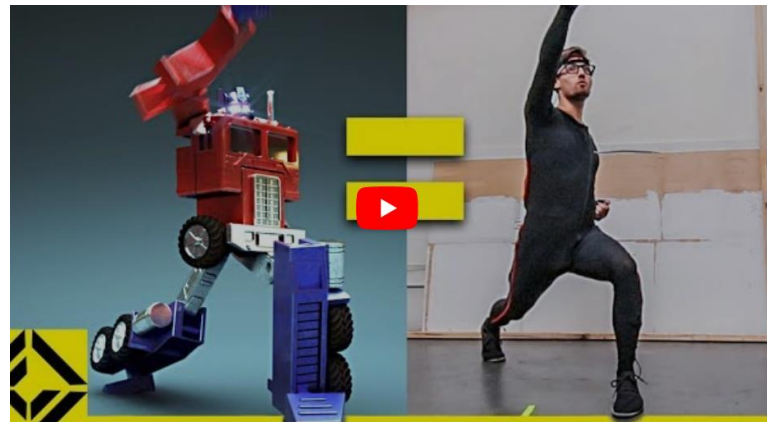
Components of motion capture

Motion capture is technology driven. This technology, however, has evolved to support actor's performances and create a better motion capture character.

1- Mocap suit

The first component of motion capture is a mocap suit. A mocap suit captures the body movements of an actor. A mocap suit has various sensors (typically 15-20) that track gravitational pull and rotation to fully capture movement.

All of this motion data is then digitally transmitted to a software to capture the motion in real time.



Examples for Mocap suits للاطلاع فقط

Rokoko Smartsuit Pro: The Rokoko Smartsuit Pro is a wireless mocap suit that uses inertial measurement units (IMUs) to track the movement of the body. It is a good option for beginners and professionals alike, and it is relatively affordable.

Perception Neuron 3: The Perception Neuron 3 is another wireless mocap suit that uses IMUs. It is more expensive than the Rokoko Smartsuit Pro, but it offers more features, such as support for gloves and facial tracking.

Xsens MVN Awinda: The Xsens MVN Awinda is a wired mocap suit that uses IMUs. It is the most expensive option on this list, but it is also the most accurate and precise.



Comparison of Mocap suits للاطلاع فقط

Feature	Rokoko Smartsuit Pro	Perception Neuron 3	Xsens MVN Awinda	Noitom Hi5 Pro	PhaseSpace Perception Neuron X1
Tracking technology	IMUs	IMUs	IMUs	Optical tracking	Optical tracking
Number of sensors	17	17	17	20	20
Wireless	Yes	Yes	No	Yes	Yes
Price	\$2,999	\$3,999	\$7,999	\$3,999	\$4,999
Accuracy	Good	Excellent	Excellent	Good	Excellent
Precision	Good	Excellent	Excellent	Good	Excellent
Ease of use	Easy	Easy	Moderate	Easy	Easy
Compatibility with software	Wide range	Wide range	Wide range	Wide range	Wide range
Warranty	1 year	1 year	2 years	1 year	1 year

Components of motion capture-2

2- Head mounted camera

- One of the more astonishing parts of motion capture is its ability to capture the details in an actor's face.
- Emotional details in the subtle ways an actor's lip quivers or eyes shift are all captured and translated to a CGI character.
- The cameras can be either **optical** or **infrared**. كاميرا عادية أو أشعة تحت الحمراء (للتصوير الليلي)
- Also, **Sensors** can be added to actor's face to capture **facial expressions**



common hardware used in motion capture

- **Markers:** Markers are **small, reflective objects** that are attached to the body or object being tracked. The markers are **tracked by the cameras** and their positions are used to reconstruct the movement of the body or object.
- **Face Markers:** Facial markers are small, reflective objects that are attached to the face. The markers are tracked by the cameras and their positions are used to reconstruct the movement of the face.

Additional hardware for **facial capture**:

- **Optical motion capture systems:** These systems use optical cameras to track the movement of markers. The markers are typically made of retroreflective material, which means that they reflect light back to the camera in the same direction that it came from. This makes it possible for the cameras to track the markers accurately, even in low-light conditions.
- **Infrared motion capture systems:** These systems use infrared cameras to track the movement of markers. Infrared light is invisible to the human eye, so it can be used in situations where visible light would be distracting or interfering.
- **Markerless motion capture systems:** These systems do not use markers. Instead, they track the movement of the body or object using computer vision techniques. This can be more challenging than using markers, but it can also be more flexible, as it does not require the subject to be wearing anything special.

Examples for Hardware in Motion Capture

للاطلاع فقط

- **Optical motion capture systems:**

- **Vicon:** Vicon is a leading provider of optical motion capture systems. Their systems are used in a variety of industries, including film, video games, and sports.
- **Motion Analysis Corporation (MAC):** MAC is another leading provider of optical motion capture systems. Their systems are used in a variety of medical applications, as well as in the film and video game industries.
- **Qualisys:** Qualisys is a Swedish company that provides optical motion capture systems. Their systems are used in a variety of industries, including sports, biomechanics, and entertainment.

- **Infrared motion capture systems:**

- **Perception Neuron:** Perception Neuron is a company that provides infrared motion capture systems. Their systems are used in a variety of applications, including virtual reality, augmented reality, and sports.
- **OptiTrack:** OptiTrack is a leading provider of infrared motion capture systems. Their systems are used in a variety of industries, including film, video games, and sports.
- **NaturalPoint:** NaturalPoint is a company that provides infrared motion capture systems. Their systems are used in a variety of applications, including sports, biomechanics, and entertainment.

Examples for Hardware in Motion Capture

2- للاطلاع فقط

- **Markerless motion capture systems:**

- **Kinect**: Kinect is a motion capture system that was developed by Microsoft. It uses a depth sensor to track the movement of the body without the need for markers.
- **Leap Motion**: Leap Motion is a motion capture system that uses infrared sensors to track the movement of the hands and fingers.
- **Faceware**: Faceware is a company that provides markerless facial capture systems. Their systems are used in a variety of applications, including film, video games, and virtual reality.

- **Head-mounted cameras:**

- **GoPro**: GoPro is a company that provides action cameras that can be mounted on a helmet or headband. These cameras can be used to capture facial expressions in three dimensions.
- **Intel RealSense**: Intel RealSense is a company that provides depth cameras that can be used to capture facial expressions in three dimensions.
- **Noitom**: Noitom is a company that provides head-mounted cameras that are specifically designed for facial capture.

Components of motion capture-3

3- Software

After shooting, the VFX team would use the motion capture data from an actor's facial expressions and performance and add details to create a fully realized digital character. To achieve this, VFX teams utilize a software.

Most commonly they are softwares made by Autodesk such as **Autodesk MotionBuilder** or 3DS Max.

للاطلاع فقط Commercial motion capture software

Autodesk MotionBuilder: MotionBuilder is a powerful motion capture software tool that is used by many professional animators and game developers. It offers a wide range of features, including support for a variety of hardware, real-time playback, and animation tools.

<https://www.autodesk.com/products/motionbuilder/overview>

C Motion: C Motion is another popular commercial motion capture software tool. It is known for its accuracy and precision, and it is often used in the film and television industry.

<https://www.rokoko.com/insights/top-8-most-popular-motion-capture-and-animation-software>

NaturalPoint Motive: Motive is a motion capture software tool that is specifically designed for use with the NaturalPoint OptiTrack system. It offers a user-friendly interface and a variety of features, including real-time playback and editing. <https://optitrack.com/software/motive/>

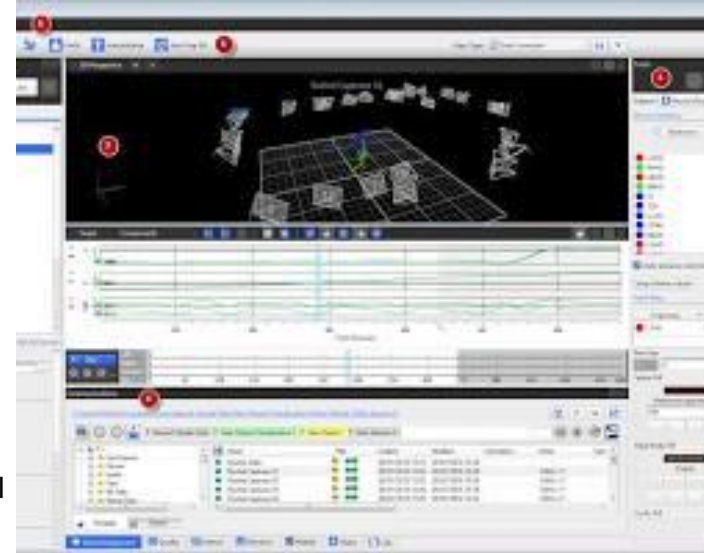


Open source motion capture software: **للاطلاع فقط**

FreeMoCap: FreeMoCap is a free and open-source motion capture software tool. It is a good option for beginners and hobbyists, as it is easy to use and does not require any special hardware. <https://www.3dart.it/en/download-free-mocap-moves-rigged-characters/>

MocapLab: MocapLab is another free and open-source motion capture software tool. It is more powerful than FreeMoCap, and it offers a wider range of features, including support for a variety of hardware and real-time playback. <https://variety.com/2022/artisans/global/motion-capture-mocaplab-1235161331/>

Vicon Nexus: Vicon Nexus is a free motion capture software tool that is provided by Vicon. It is a good option for users who already have a Vicon motion capture system, as it allows them to view and analyze their data without having to purchase additional software. <https://docs.vicon.com/pages/viewpage.action?pagelId=50888731>

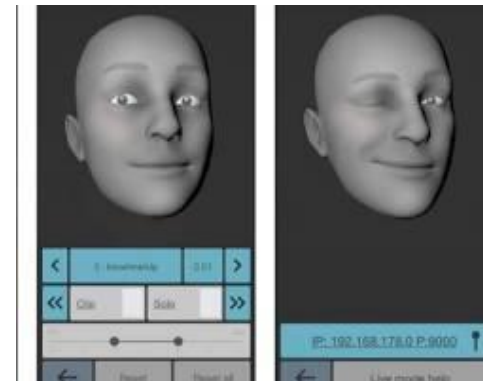


Commercial facial capture software: للاطلاع فقط

Faceware Studio: Faceware Studio is a powerful facial capture software tool that is used by many professional animators and game developers. It offers a wide range of features, including support for a variety of hardware, real-time playback, and animation tools. <https://facewaretech.com/>

iClone 7 Character Creator: iClone 7 Character Creator is a facial capture software tool that is specifically designed for use with the iClone 7 animation software. It offers a user-friendly interface and a variety of features, including real-time playback and editing. <https://mocap.reallusion.com/iClone-faceware-mocap/>

Face Cap: Face Cap is a facial capture software tool that is specifically designed for use with the Unity game engine. It offers a variety of features, including real-time playback and editing, as well as the ability to export your data to other software. <https://www.iphoneness.com/iphone-apps/face-cap/>



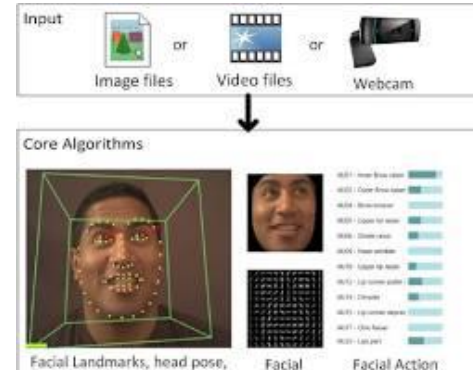
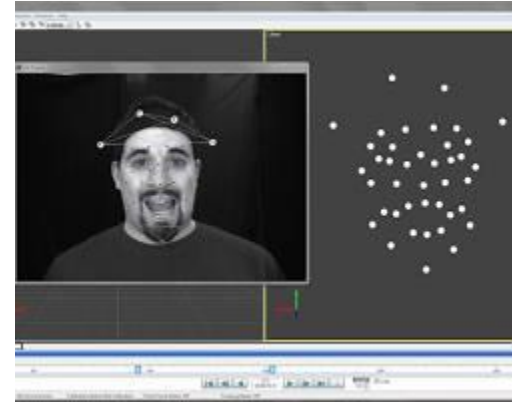
Open source facial capture software: للاطلاع فقط

FaceTracker: FaceTracker is a free and open-source facial capture software tool. It is a good option for beginners and hobbyists, as it is easy to use and does not require any special hardware.

<https://optitrack.com/software/expression/indepth.html>

ZED SDK: ZED SDK is a free and open-source facial capture software tool that is specifically designed for use with the ZED stereo camera. It offers a variety of features, including real-time playback and editing, as well as the ability to export your data to other software.

OpenFace: OpenFace is a free and open-source facial capture software tool that is specifically designed for research purposes. It offers a variety of features, including face detection, facial landmark detection, and facial expression recognition.



What is **compositing**?

- **Compositing** is the process through which two or more images combine to make the appearance of a single picture. The composite process can be done on-set and in-camera or during Post-Production.
- There are dozens of different ways to composite shots but perhaps the most common example is when a weatherman is placed in front of a greenscreen with the weather details behind them.
- تركيب مشاهد متعددة لتبدو كأنها مشهد موحد

1-Rotoscope animation

- Rotoscope animation is the process through which live action **footage** المشهد الأساسي is **traced frame-by-frame**.
تتبع مشهد لقطة بلقطة مع اضافة تفاصيل يدوياً
- A Rotoscope effect can also be achieved in live action films by matting تحديد حصر out objects – like it was here in the original *Star Wars*.



2-Chroma keying

- Chroma keying is essentially the process by which a color is removed from an image.
- For example: it's the process by which green is removed from a green screen video and replaced with something else in Post-Production.



3-Double Exposure

- **Double exposure** is a technique that **combines** two different exposures or images that are **layered** on top of each other. The image overlaid is **less than full opacity** so a bit of **both images can be seen** producing an almost ghost-like image.
- In cinema, double exposure can convey many emotions — especially films of the cerebral or even nefarious kind



4-Bullet time

- Bullet time was a favorite VFX technique used in filmmaking. It gives a **slow-motion** effect and shows the **complex** details of an action.



5-Doubling

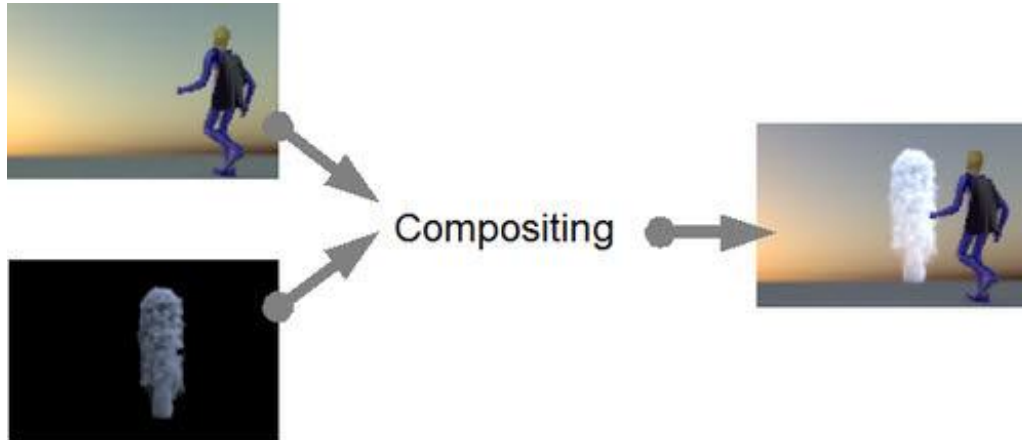
- The act of doubling involves projecting **twins** التوأم or clones.
- In reality, an actor who doesn't have a twin is filmed individually at different locations, and the resultant footage is combined to project twins or clones.



Case Study:

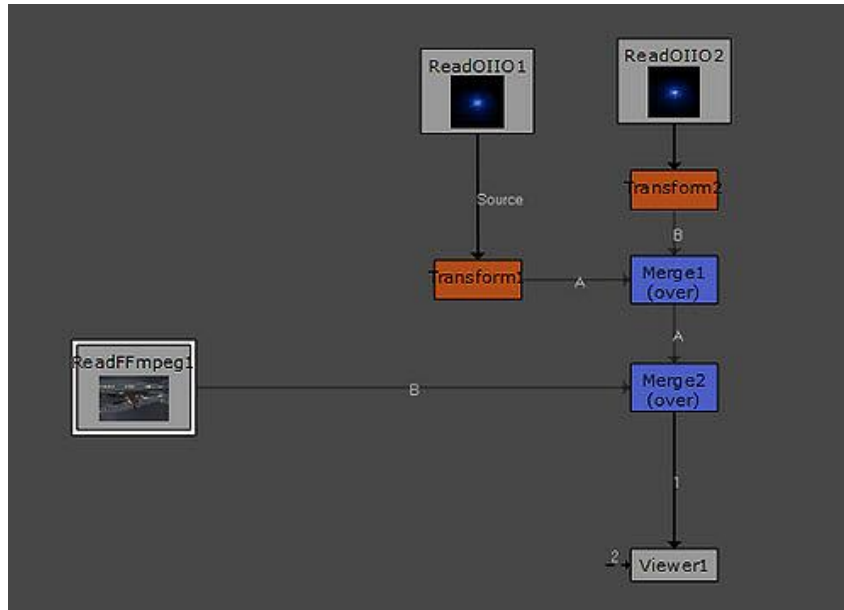
Introduction to Natron for Digital Compositing

- Natron is a node-based digital compositor. Composite Nodes are individual blocks that perform a certain operation. Compositing Nodes allow you to take multiple renders, videos, images and other sources as inputs, and process a variety of operations on them to combine them in different ways.



Case Study:Node

- Node setups will look like blocks connected by lines (arrows). Below shows an example of a Node configuration



Natron Nodes Rules

1. Nodes can have **1 or more inputs**.

Most processing nodes have only 1 input (blur, color corrections...)

Merging nodes have several inputs that are turned into one “mixed” output

2. **mask input is present on many nodes.**

It's purpose usually is to **limit** the effect of the node to the part of the image defined by the white part of the mask

3. Nodes always have **exactly 1 output**.

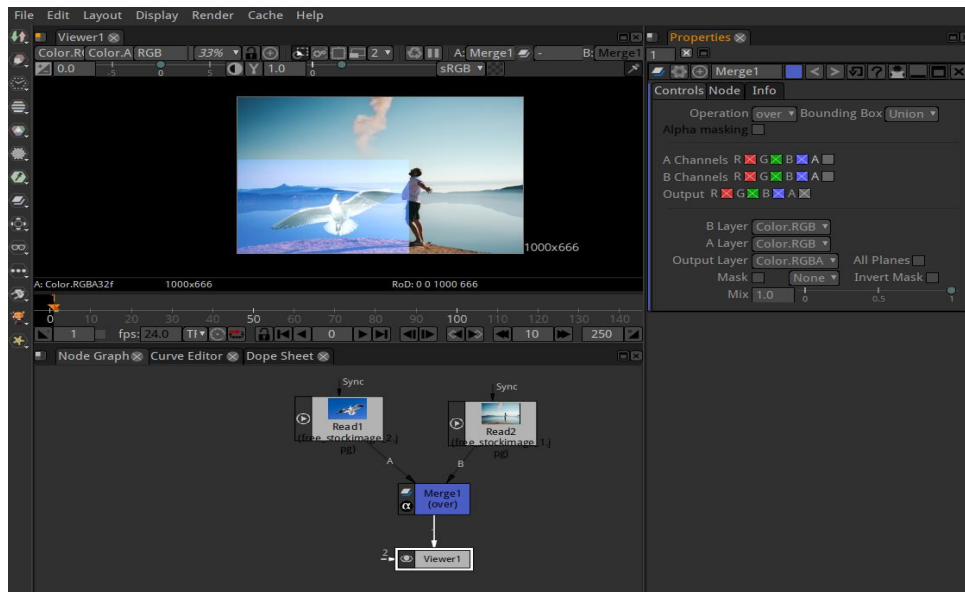
If several nodes B and C connect their inputs to the same output of node A they will receive the same image

The only node without any output is the Viewer node. It determines what node is shown in the Viewer pane



Merging images-1

- With the merge node you are able to control how your images are combined.
- **Layering Images Together with the Merge Node**
 1. Select Merge > Merge (or press M in the Node Graph) to insert a **Merge node** after the images you want to layer together.
 2. Connect your images to the Merge node's **A** and **B** inputs.
 3. Connect a Viewer to the output of the Merge node so you can see the effect of your merge operation.



Merging images-2

4-In the Merge node's controls, select how you want to layer the images together from the operation dropdown menu.

The **most common operation is over**, which layers input **A** over input **B** according to the **alpha of input A**.



Chroma Key Example

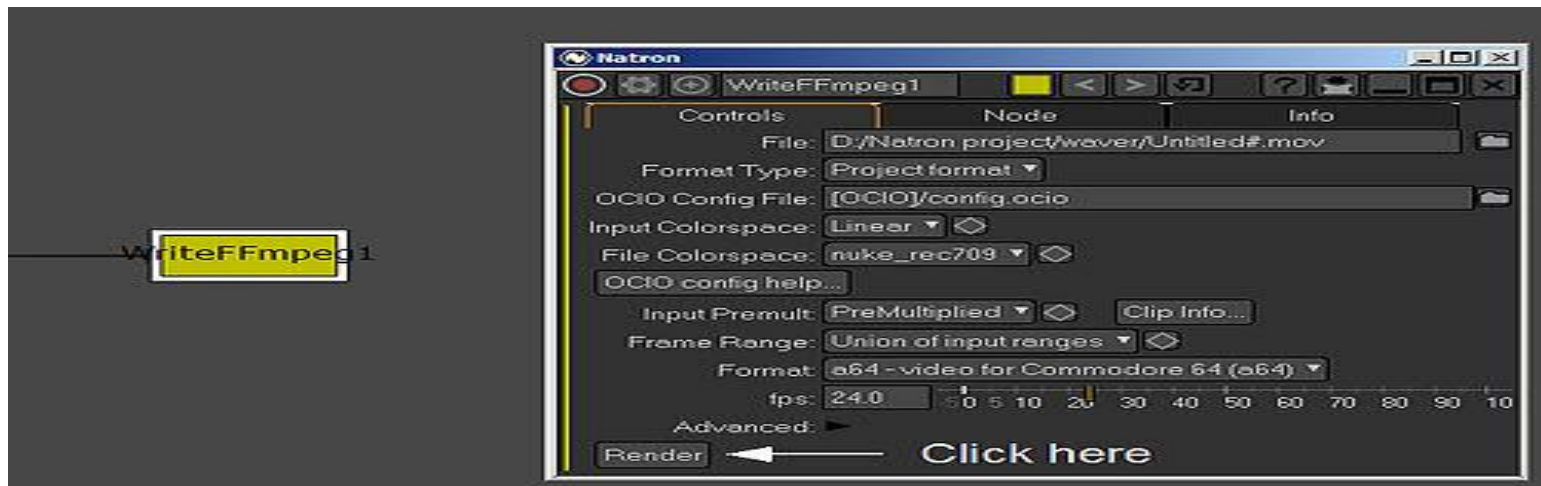
1. Add a green screen shot with a **Read node** (r on your keyboard, or a right-click anywhere in the Node Graph window).
2. Add a background shot or matte photo, also with a Read node (r on your keyboard, or a right-click anywhere in the Node Graph window).
3. Add the "green screen effect", more technically known as a **chroma key**, from Keyer > ChromaKeyer.
4. **Connect** the **green screen** node to the **chromakeyer** source.
5. **Connect** the **background** node to the **chromakeyer** background.
6. **Connect** the output of the Chromakeyer node to the **Viewer** node.



<https://opensource.com/life/15/7/getting-started-with-natron>

Rendering final output to file (video)

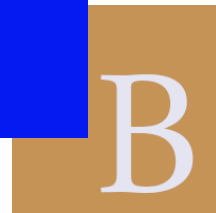
- in order to render the final result to a file, a **Write Node** is required in the setup.
- When it is time to do the render to file, click on the Render button corresponding to the Write Node. This is as indicated below.
- This will render a file to be saved in the assigned location.



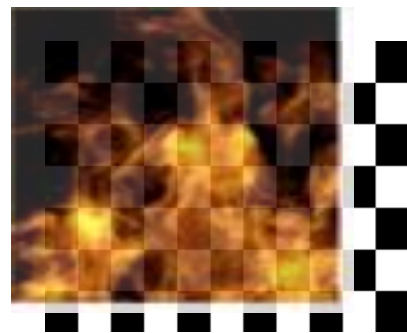
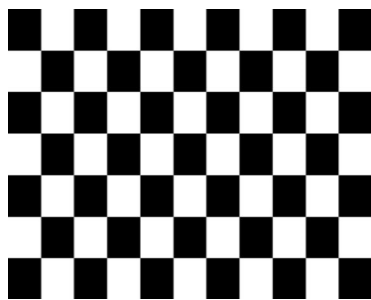
Merge Operations

- There are many merge operations
- In the following slides, we shall check some operations
 - Each operation will be applied on 2 samples (Each with 2 images)
 - Sample 1: one that combines the letters A and B to a merged image
 - Sample 2: that has an image of fire merged with the familiar checkerboard

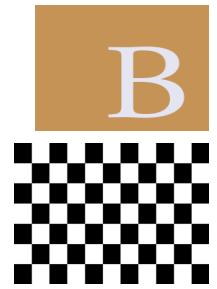
Sample 1 images



Sample 2 images



Merge Operations-1



1- over:

This is the **default** operation. Layers image A over B according to the alpha of image A.

Sample 1

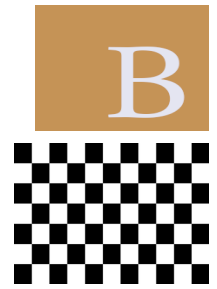


Sample 2



https://learn.foundry.com/nuke/13.2v4/content/comp_environment/merging/merge_operations.html

Merge Operations-2



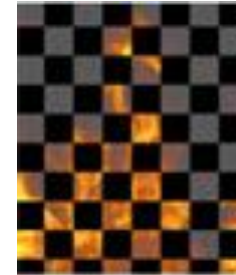
2- Atop:

Shows the shape of image B, with **A covering B** where the images **overlap** في مناطق التقاطع.

Sample 1



Sample 2



https://learn.foundry.com/nuke/13.2v4/content/comp_environment/merging/merge_operations.html

Merge Operations-3



3- **average**

The average of the two images. The result is **darker** than the original images.

Sample 1



Sample 2

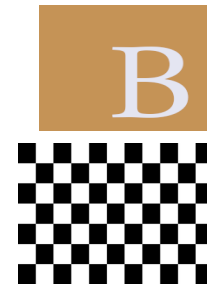


https://learn.foundry.com/nuke/13.2v4/content/comp_environment/merging/merge_operations.html

Merge Operations-4

4- copy

- Only shows image A.



Sample 1

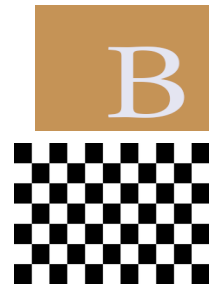


Sample 2



https://learn.foundry.com/nuke/13.2v4/content/comp_environment/merging/merge_operations.html

Merge Operations-4



5- difference

- How much the pixels differ.
- Useful for **comparing two very similar images**. This mode can also be used as a **difference keyer**. مقارنة لتوضيح الفروق

Sample 1



Sample 2



https://learn.foundry.com/nuke/13.2v4/content/comp_environment/merging/merge_operations.html

Merge Operations-5

5- from (B-A)

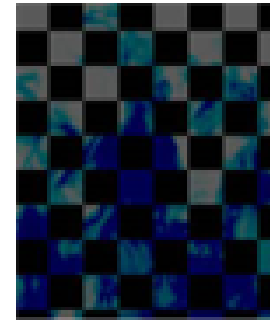
Image A is subtracted from B.



Sample 1

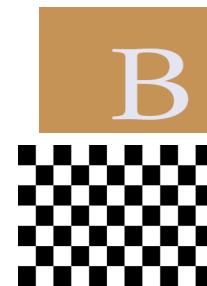


Sample 2



https://learn.foundry.com/nuke/13.2v4/content/comp_environment/merging/merge_operations.html

Merge Operations-6



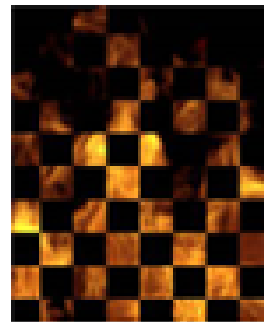
6- in: التقاطع

Only shows the areas of image A that overlap with the alpha of B

Sample 1

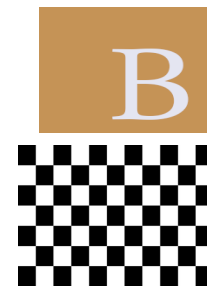


Sample 2



https://learn.foundry.com/nuke/13.2v4/content/comp_environment/merging/merge_operations.html

Merge Operations-7



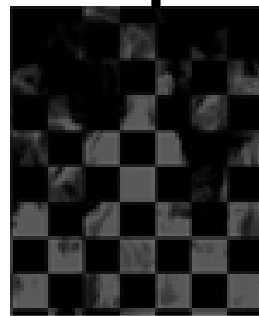
7- mask:

This is the **reverse** of the (in) operation. Only shows the areas of image B that overlap with the alpha of A (visible parts of A) = (A is Mask for B)

Sample 1



Sample 2

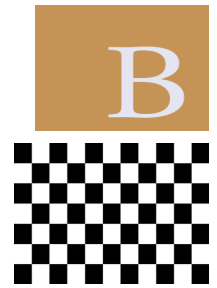


https://learn.foundry.com/nuke/13.2v4/content/comp_environment/merging/merge_operations.html

Merge Operations-8

8- max:

Takes the **maximum** values of both images



Sample 1

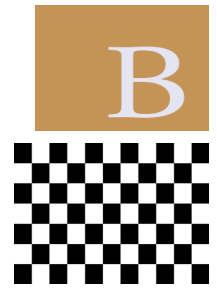


Sample 2



https://learn.foundry.com/nuke/13.2v4/content/comp_environment/merging/merge_operations.html

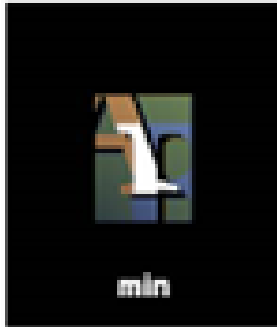
Merge Operations-9



9- min :

Takes the minimum values of both images

Sample 1



Sample 2

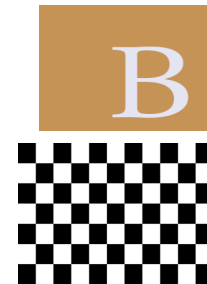


https://learn.foundry.com/nuke/13.2v4/content/comp_environment/merging/merge_operations.html

Merge Operations-10

10-color-burn

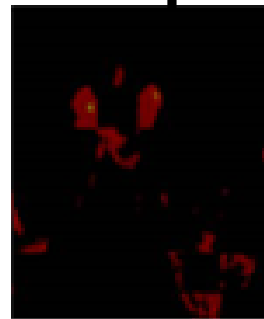
يغمق الصورة الثانية كي تصل للأولى darken B towards A



Sample 1



Sample 2

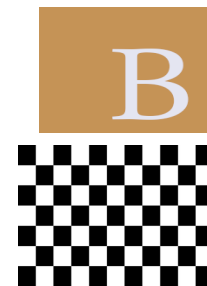


https://learn.foundry.com/nuke/13.2v4/content/comp_environment/merging/merge_operations.html

Merge Operations-11

11-xor

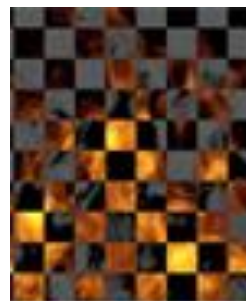
Shows both image A and B where the images do not overlap.



Sample 1



Sample 2



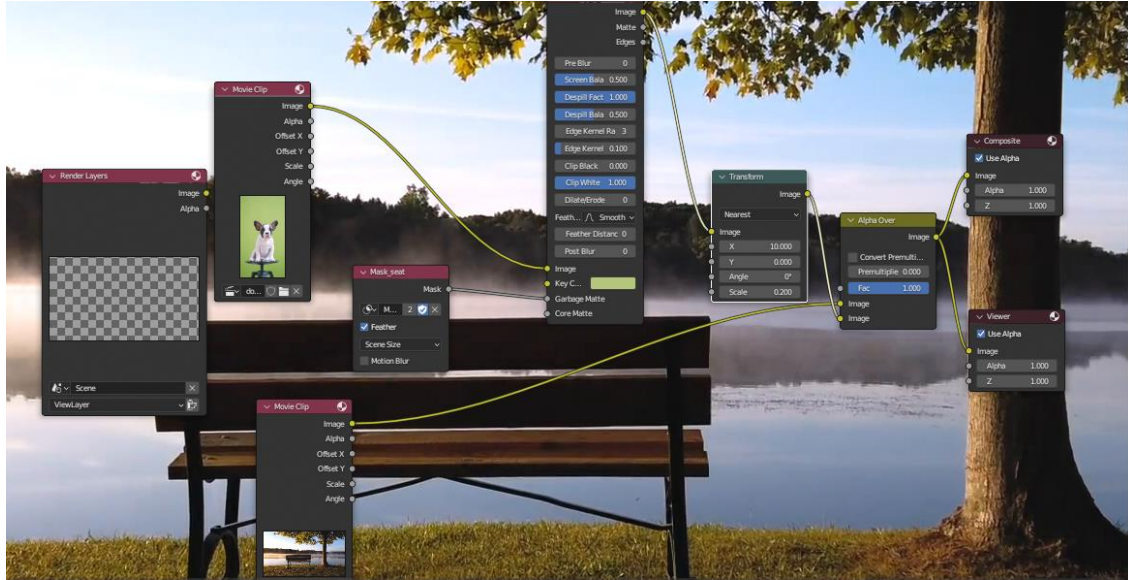
https://learn.foundry.com/nuke/13.2v4/content/comp_environment/merging/merge_operations.html

Blender GreenScreen Compositing tutorials and case study

<https://www.youtube.com/watch?v=KBBkgoDPnZE>

<https://www.youtube.com/watch?v=m5qbZNGgrcA>

<https://www.youtube.com/watch?v=IA6AuZYbLyQ>



thanks

references

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- [https://corddigital.com/Visual-Effect-\(VFX\)-impact-Nowadays-and-its-types](https://corddigital.com/Visual-Effect-(VFX)-impact-Nowadays-and-its-types)
- <https://www.nfi.edu/vfx/>
- <https://www.studiorola.com/tutorials/compositing/tutorial-basic-introduction-to-natron-for-compositing-part-1/>
- <https://opensource.com/life/15/7/getting-started-with-natron>
- https://learn.foundry.com/nuke/content/comp_environment/merging/merge_operations.html
- <https://www.bhphotovideo.com/explora/video/tips-and-solutions/filmmaking-101-camera-shot-types>